**ANTIDERIVATIVE (INTEGRATION)** Finding antiderivative (or Integration) of any function is reverse process of differentiation. Mathematically a function $F(x)$ is antiderivative of function $f(x)$ if $F'(x) = f(x)$ for the domain where $f(x)$ and $F(x)$ are defined. For example If $f(x) = x$ then if $F(x) = \frac{x^2}{2} + C$ then $F'(x)$ yields $x$, thus we say that $F(x) = \frac{x^2}{2} + C$ is the integration of $f(x) = x$.

(Here $C$ is a constant). Similarly if $f(x) = 3x^2 + 2x + 4$ then you can verify that its antiderivative is $x^3 + x^2 + 4x + C$ ($C$ could be any arbitrary constant).

**EXAMPLE:** Find the anti-derivative of each of the following functions:
(a) $f(x) = 2x$ (b) $g(x) = \cos(x)$ (c) $h(x) = \frac{1}{x} + 2e^{2x}$

**Solution:** Try to think what functions on differentiation will give the required $f(x)$. Thus:
(a) $F(x) = x^2 + C$ (b) $G(x) = \sin(x) + C$ (c) $H(x) = \ln|x| + e^{2x} + C$

→ Note that $x^2$ is not the only function whose derivative is $2x$ $x^2+1$, $x^2+2$ etc all have a derivative $2x$ thus the required solution involves an arbitrary constant.

**EXAMPLE:** Find anti-derivative of $f(x) = 3x^2$ that satisfies $F(1) = -1$
**Solution:** $F(x) = x^3 + C$, now $F(1) = 1 + C = -1 \Rightarrow C = -2$ thus our required function, $F(x) = x^3 - 2$ Ans.

**NOTATION FOR ANTIDERIVATIVE (INTEGRATION):**

$F(x) = \int f(x)\,dx$ is read as integration of $f(x)$ wrt $x$ or

alternatively as Antiderivative of $f(x)$ with respect to $x$.

In the coming sections we will be learning the techniques and rules for finding integrations for various functions.

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# TABLE OF STANDARD ANTI-DERIVATIVES:

| Function | General antiderivative | Function | General antiderivative |
|---|---|---|---|
| 1. $x^n$ | $\frac{1}{n+1}x^{n+1} + C, \quad n \neq -1$ | 8. $e^{kx}$ | $\frac{1}{k}e^{kx} + C$ |
| 2. $\sin kx$ | $-\frac{1}{k}\cos kx + C$ | 9. $\frac{1}{x}$ | $\ln \lvert x \rvert + C, \quad x \neq 0$ |
| 3. $\cos kx$ | $\frac{1}{k}\sin kx + C$ | 10. $\frac{1}{\sqrt{1-k^2x^2}}$ | $\frac{1}{k}\sin^{-1} kx + C$ |
| 4. $\sec^2 kx$ | $\frac{1}{k}\tan kx + C$ | 11. $\frac{1}{1+k^2x^2}$ | $\frac{1}{k}\tan^{-1} kx + C$ |
| 5. $\csc^2 kx$ | $-\frac{1}{k}\cot kx + C$ | 12. $\frac{1}{x\sqrt{k^2x^2-1}}$ | $\sec^{-1} kx + C, kx > 1$ |
| 6. $\sec kx \tan kx$ | $\frac{1}{k}\sec kx + C$ | 13. $a^{kx}$ | $\left(\frac{1}{k \ln a}\right)a^{kx} + C, a > 0, a \neq 1$ |
| 7. $\csc kx \cot kx$ | $-\frac{1}{k}\csc kx + C$ | | |

→ Students should by heart memorize the table of antiderivatives because these will be repeatedly required for integrating various functions.

→ $\int f(ax+b) = \frac{1}{a}\int f(x)dx$ where $x = ax+b$ for example,

$\int (3x+4)^2 dx = \frac{1}{3}\left(\frac{x^3}{3}\right)$ where $x = (3x+4)$ thus substituting

x back as ax+b: $\int (3x+4)^2 = \frac{1}{9}(3x+4)^3 + C$. Student

can readily differentiate is back to get $(3x+4)^2$.

$\frac{d}{dx}\left[\frac{1}{9}(3x+4)^3\right] = \frac{1}{9} \times 3(3x+4)^2 \times 3 = (3x+4)^2$.

**EXAMPLE:** Find antiderivatives of (a) $(3x+4)^3$ (b) $\cos(3x+4)$ (c) $\frac{1}{3x+4}$ (d) $e^{3x+4}$

**Solution:** Using our last rule, our answers are as follows:

(a) $\frac{1}{3}\frac{(3x+4)^4}{4} + C$ $(\because X = 3x+4)$

(b) $\frac{1}{3}\sin(3x+4) + C$ $(\because X = 3x+4)$

(c) $\frac{1}{3}\ln\lvert 3x+4 \rvert + C$ $(\because X = 3x+4)$

(d) $\frac{1}{3}e^{3x+4} + C$ $(\because X = 3x+4)$

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**EXAMPLE:** Using the table of antiderivatives find anti-derivatives of following functions: (a) $x^5$ (b) $1/\sqrt{x}$ (c) $\sin 2x$ (d) $\cos(x/2)$ (e) $e^{3x}$ (f) $2^x$.

**Solution:** (a) $\frac{1}{6}x^6 + C$ (Formula 1, $n=5$) (b) $\frac{x^{1/2}}{1/2} + C = 2\sqrt{x} + C$

(∵ Formula 1 with $n = -1/2$) (c) $-\frac{1}{2}\cos(2x) + C$ (∵ Formula 2 with $b=2$)

(d) $2\sin(x/2) + C$ (∵ Formula 3 with $b = 1/2$) (e) $\frac{-1}{3}e^{-3x} + C$ (∵ Formula 8 with $b = -3$) (f) $\frac{1}{\ln(2)}2^x + C$ (∵ formula 13 with $a=2, k=1$).

**ANTIDERIVATIVE LINEARITY RULES:**

| | Function | General antiderivative |
|---|---|---|
| 1. *Constant Multiple Rule:* | $kf(x)$ | $kF(x) + C$, $k$ a constant |
| 2. *Sum or Difference Rule:* | $f(x) \pm g(x)$ | $F(x) \pm G(x) + C$ |

**EXAMPLE:** Find general antiderivative of $f(x) = \frac{3}{\sqrt{x}} + \sin(2x)$.

**Solution:** Using constant multiple and sum rule:

$$F(x) = \int \frac{3}{\sqrt{x}} + \sin(2x)\,dx = 3\int \frac{1}{\sqrt{x}}\,dx + \int \sin(2x)\,dx$$

thus $F(x) = 3 \times 2\sqrt{x} - \frac{1}{2}\cos(2x) + C = 6\sqrt{x} - \frac{1}{2}\cos(2x) + C$ ans.

**DIFFERENTIAL EQUATIONS:** A relationship between various derivatives of a function is called a differential equation and a function which makes the differential equation true is called a solution to the differential equation. For instance:

$y''^2 + 2y' + y = x^2 + 4x + 4$ ① is a second order non linear differential equation (Second order because highest derivative is the second derivative, non linear because derivatives have powers other than 1 (y can be considered as zeroth derivative of itself)).

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For differential equation (1), student can readily verify that $y = x^2$ is one of its solutions.

**VARIABLE SEPARABLE DIFFERENTIAL EQUATIONS:** A differential equation of the form $\frac{dy}{dx} = \frac{f(x)}{g(y)}$ is called a variable separable differential equation.

A variable separable differential equation is solved by separating the variables! i.e. writing the equation successively as:

$$g(y)dy = f(x)dx \quad \text{and then:}$$

$$\int g(y)dy = \int f(x)dx.$$

**EXAMPLE:** Solve the differential equation $\frac{dy}{dx} = \frac{e^x}{y}$.

**Solution:** $\frac{dy}{dx} = \frac{e^x}{y} \Rightarrow y\,dy = e^x dx \Rightarrow \int y\,dy = \int e^x dx$

$$\Rightarrow \frac{y^2}{2} = e^x + C \quad \text{or} \quad y = \pm\sqrt{2e^x + C}\ dx.$$

**INITIAL VALUE PROBLEMS:** In the differential equation that we just solved, there was an arbitrary constant, however, differential equations are often used to describe physical systems, which have a particular response depending on initial conditions at the time of starting the system. An equation describing such a system cannot have any arbitrary constant and the constant has to be evaluated to satisfy the initial conditions of the system. Such a solution is called particular solution to the initial value problem.

**EXAMPLE:** Solve the initial value problem: $\frac{d^2s}{dt^2} = -k$, Initial conditions, $\frac{ds}{dt} = 88$ and $s = 0$ when $t = 0$.

**Solution:** $\frac{d^2s}{dt^2} = -k$, now integrating with respect to time,

$$\frac{ds}{dt} = -kt + C \quad \text{now at } t = 0,\ \frac{ds}{dt} = 88 \Rightarrow -k \times 0 + C = 88 \Rightarrow C = 88$$

thus, $\frac{ds}{dt} = -kt + 88$, now integrating once again,

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$s = -\frac{1}{2}ht^2 + 88t + C$, now at $t=0$, $s=0 \Rightarrow C=0$

Thus $s = -\frac{1}{2}ht^2 + 88t$ Ans.

**EXAMPLE:** Solve the initial value problem $\frac{dr}{d\theta} = -\pi \sin \pi\theta$, $r(0)=0$.

**Solution:** $dr = -\pi \sin \pi\theta \, d\theta \Rightarrow \int r^0 dr = -\pi \int \sin \pi\theta \, d\theta$

$\Rightarrow r = -\pi \times \frac{1}{\pi}(-\cos \pi\theta) + C = \cos \pi\theta + C$ now @ $\theta=0$, $r=0$

$\Rightarrow 0 = \cos 0 + C \Rightarrow C=-1 \Rightarrow r = \cos \pi\theta - 1$ Ans.

**EXAMPLE:** Solve the initial value problem $\frac{d^2s}{dt^2} = \frac{3t}{8}$, $\left.\frac{ds}{dt}\right|_{t=4} = 3$, $s(4)=4$

**Solution:** $\frac{d^2s}{dt^2} = \frac{3t}{8}$, Integrating both sides wrt $t$:

$$\frac{ds}{dt} = \frac{3t^2}{16} + C \quad \text{now using } \frac{ds}{dt} = 3 \text{ @ } t=4,$$

$3 = \frac{3}{16} \times 4^2 + C \Rightarrow C=0 \Rightarrow \frac{ds}{dt} = \frac{3t^2}{16}$, now integrating wrt $t$:

$$s = \frac{t^3}{16} + C \quad \text{now using } s=4 \text{ @ } t=4, \; 4 = \frac{4^3}{16} + C \Rightarrow C=0$$

$\Rightarrow s = t^3/16$ Ans.

**EXAMPLE:** Solve the initial value problem $\frac{dv}{dt} = \frac{8}{1+t^2} + \sec^2(t)$, $v(0)=1$

**Solution:** Integrating wrt $t$:

$$v = 8\tan^{-1}(t) + \tan(t) + C \quad \text{Now @ } t=0, \; v=1$$

$\Rightarrow 1 = 0 + 0 + C \Rightarrow C=1 \Rightarrow v = 8\tan^{-1}(t) + \tan(t) + 1$ Ans.

**EXAMPLE:** Solve the initial value problem $\frac{dy}{dx} = \frac{x^2}{y}$, $y(0)=3$

**Solution:** Separating the variables $y\,dy = x^2 dx \Rightarrow \int y\,dy = \int x^2 dx$

thus $y^2/2 = x^3/3 + C$ now at $x=0$, $y=3$ thus:

$9/2 = 0/3 + C \Rightarrow C = 4.5$ thus solution :

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$y^2/2 = x^3/3 + 4.5$ Ans.

**DEFINITE INTEGRAL:** Consider the following notation:

$$\int_a^b f(x)\,dx$$

This is read as "definite integration of $f(x)$ wrt $x$ from $a$ to $b$"

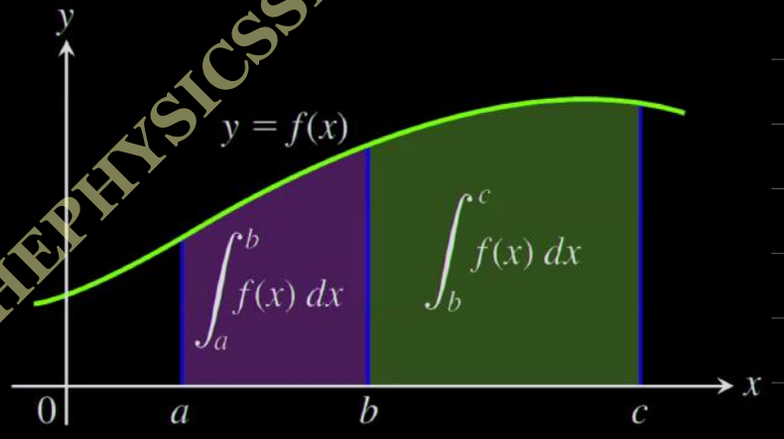
If $F(x)$ is the antiderivative of $f(x)$ then we define:

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Geometrically it represents area under the graph $y = f(x)$ from $x = a$ to $x = b$

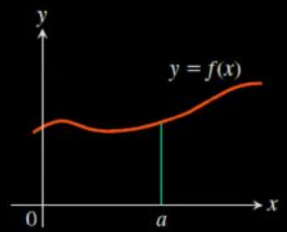
→ By area enclosed under the graph we mean the area enclosed between the curve and $x$ axis

→ This area is considered as a signed value. It is positive if the curve lies above $x$ axis and negative if curve lies below $x$-axis.



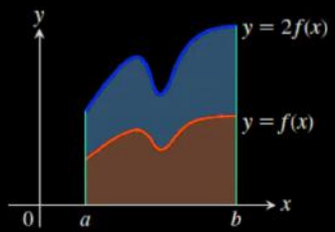

→ $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$, this follows directly from definition of definite Integral. Effectively this says, $F(b) - F(a) = -(F(a) - F(b))$ which is obviously true.

**PROPERTIES OF DEFINITE INTEGRAL:**



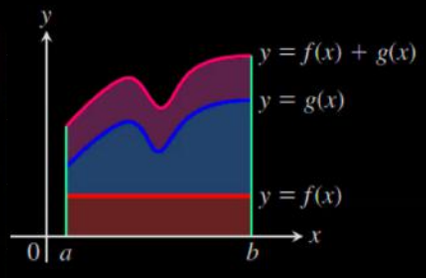

(a) *Zero Width Interval:*

$$\int_a^a f(x)\,dx = 0$$




(b) *Constant Multiple:* $(k = 2)$

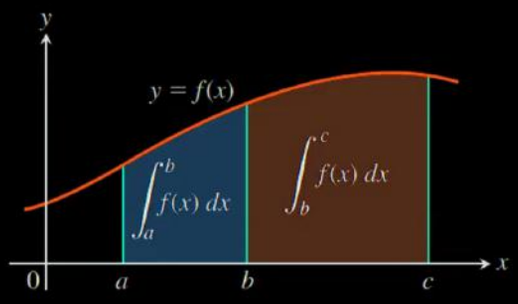
$$\int_a^b kf(x)\,dx = k\int_a^b f(x)\,dx$$




(c) *Sum: (areas add)*

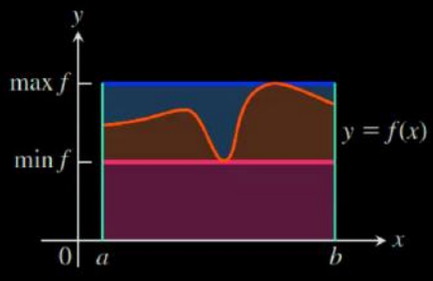
$$\int_a^b (f(x) + g(x))\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$$

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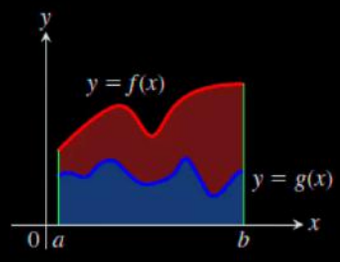

(d) *Additivity for Definite Integrals:*

$$\int_a^b f(x)\,dx + \int_b^c f(x)\,dx = \int_a^c f(x)\,dx$$




(e) *Max-Min Inequality:*

$$\min f \cdot (b-a) \le \int_a^b f(x)\,dx \le \max f \cdot (b-a)$$




(f) *Domination:*

$f(x) \ge g(x)$ on $[a,b]$

$$\Rightarrow \int_a^b f(x)\,dx \ge \int_a^b g(x)\,dx$$

**EXAMPLE:** Show that value of $\int_0^1 \sqrt{1+\cos x}\,dx$ is less than or equal to $\sqrt{2}$.

**Solution:** We know that $\cos x \le 1$, thus $f(x) = \sqrt{1+\cos x} \le \sqrt{2} = g(x)$

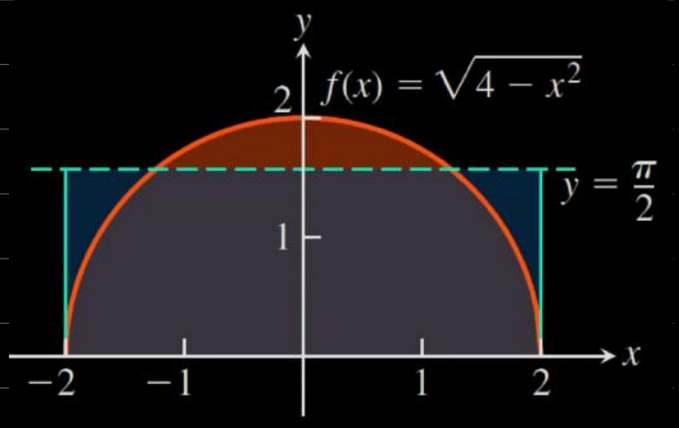
Thus $\int f(x) \le \int g(x) = \int_0^1 \sqrt{2}\,dx = \sqrt{2}$. Thus $\int f(x) \le \sqrt{2}$ QED.

**AVERAGE VALUE OF A CONTINUOUS FUNCTION:** If $f(x)$ is a continuous function on the interval $(a,b)$ then average value of $f(x)$ on the interval is defined as:

$$\langle y \rangle = \langle f(x) \rangle = \frac{1}{b-a} \int_a^b f(x)\,dx.$$

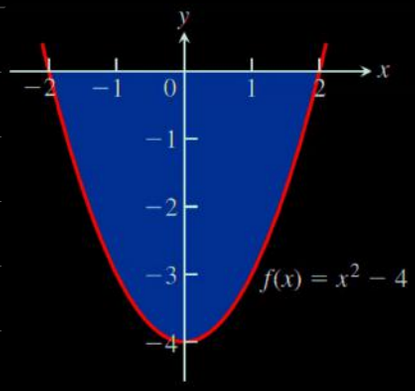
graphically it represents a function of constant height, under which the area is same as that under the given function.

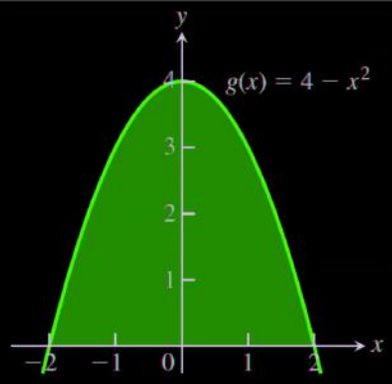
For instance consider the adjacent figure, here area under the dotted line (i.e $y = \pi/2$) is same as that under $y = f(x) = \sqrt{4-x^2}$, thus average value of $\sqrt{4-x^2}$ between $x = -2$ and 2 is $\pi/2$.




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**EXAMPLE:** The figure shows graph of functions $f(x) = x^2 - 4$ and its mirror image $g(x) = 4 - x^2$ reflected across x-axis, find:
(a) Definite integral over $[-2, 2]$
(b) Area between the graph and x-axis over this interval.




**Solution:** (a) $\int f(x)\,dx = \left[\frac{x^3}{3} - 4x\right]_{-2}^{2} = \left(\frac{8}{3} - 8\right) - \left(-\frac{8}{3} + 8\right) = -32/3$

Similarly $\int g(x)\,dx = \left[4x - \frac{x^3}{3}\right]_{-2}^{2} = +32/3.$

(b) Area in both cases has magnitude 32/3 (if area is considered as signed value).

**INTEGRATION BY SUBSTITUTION:** Many times we are not able to see a function as directly integrable from the table, however by carefully observing the function we might be able to find a suitable substitution to convert the integration in to a standard recognizable form. This will be clear from following examples:

**EXAMPLE:** Find $\int (x^3 + x)(3x^2 + 1)\,dx$

**Solution:** Recognize that $3x^2 + 1$ is derivative of $x^3 + x$. This suggests that we can try the substitution $u = x^3 + x \Rightarrow \frac{du}{dx} = 3x^2 + 1 \Rightarrow (3x^2 + 1)\,dx = du$.

This transforms our integration in to $\int u\,du = \frac{u^2}{2} + C = \frac{(x^3 + x)^2}{2} + C$ Ans.

**EXAMPLE:** Find $\int x^2 \cos(x^3)\,dx$

**Solution:** Recognize that $3x^2$ is derivative of $x^3$, this suggests we use $u = x^3 \Rightarrow du = 3x^2\,dx$ thus our integration transforms to:

$$\frac{1}{3}\int \cos u\,du = \frac{1}{3}\sin(u) + C = \frac{1}{3}\sin(x^3) + C \text{ Ans.}$$

**EXAMPLE:** Find $\int x\sqrt{2x+1}\,dx$

**Solution:** Let us try the substitution $u = 2x + 1$ and see if it helps.

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$du = 2\,dx$ and $x = \left(\frac{u-1}{2}\right)$. Thus our integration becomes:

$$\frac{1}{4}\int (u-1)\sqrt{u}\,du = \frac{1}{4}\int u^{3/2} - u^{1/2}\,du = \frac{1}{4}\left(\frac{2}{5}u^{5/2} - \frac{2}{3}u^{3/2}\right) + C$$

$$= \frac{1}{10}(2x+1)^{5/2} - \frac{1}{6}(2x+1)^{3/2} + C \quad \text{Ans.}$$

## Using Trigonometric Identities:

$$\int \sin^2(x)\,dx = \int \frac{(1-\cos 2x)}{2}\,dx = \frac{x}{2} - \frac{\sin(2x)}{4} + C$$

$$\int \cos^2(x)\,dx = \int \frac{1+\cos 2x}{2}\,dx = \frac{x}{2} + \frac{1}{4}\sin(2x) + C$$

$$\int (1-2\sin^2 x)\sin(2x)\,dx = \int \cos(2x)\sin(2x)\,dx = \frac{1}{2}\int \sin(4x)\,dx$$

$$= -\frac{1}{8}\cos(4x) + C$$

**Example:** Find $\int \frac{2z\,dz}{(z^2+1)^{3/2}}$

**Solution:** Taying $u = z^2+1$, $du = 2z\,dz$. Thus, integration transforms:

$$\int u^{-3/2}\,du = -2u^{-1/2} + C = -2(z^2+1)^{-1/2} + C \quad \text{Ans.}$$

**Example:** $\int \frac{\sin(2t+1)}{\cos^2(2t+1)}\,dt$

**Solution:** Put $= u = \cos(2t+1) \Rightarrow du = -2\sin(2t+1)\,dt$, Now

$$\frac{-1}{2}\int \frac{1}{u^2}\,du = \frac{-1}{2} \times \frac{-1}{u} + C = \frac{1}{2u} + C = \frac{1}{2\cos(2t+1)} + C$$

**Example:** Evaluate $\int \frac{18\tan^2 x \sec^2 x}{(2+\tan^3 x)^2}\,dx$ use $u = 2 + \tan^3(x)$

**Solution:** $du = 3\tan^2 x \sec^2 x\,dx \Rightarrow \int \frac{6\,du}{u^2} = \frac{-6}{u} + C$

$$= \frac{-6}{2+\tan^3 x} + C \quad \text{Ans.}$$

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## Definite Integration By Substitution:

$$\int_a^b f(g(x)) \times g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du \quad \text{where } u = g(x).$$

In short, when integration variable is changed, the limits are also changed according to the substitution. This will be clear from the following examples.

**Example:** Evaluate: $\int_{-1}^{1} 3x^2\sqrt{x^3+1}\,dx$

**Solution:** Put $u(x) = x^3+1 \Rightarrow du = 3x^2dx$ and $u(-1) = 0$ and $u(1)=2$

Thus: $$\int_0^2 \sqrt{u}\,du = \frac{2}{3}\left[u^{3/2}\right]_0^2 = \frac{2}{3}\cdot 2^{3/2} \text{ Ans.}$$

**Example:** Evaluate $\int_0^{\pi/2} \frac{2\sin(x)\cos(x)}{(1+\sin^2 x)^3}\,dx$

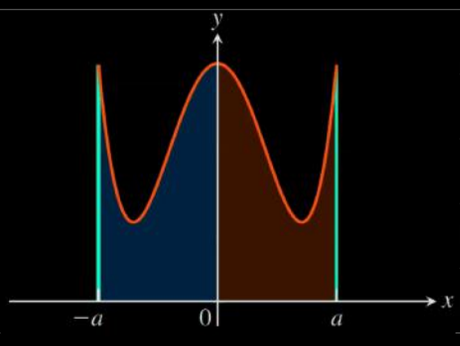
**Solution:** Put $u = 1+\sin^2 x \Rightarrow du = 2\sin(x)\cos(x)$, $u(0)=1$, $u(\pi/2)=2$

$$\Rightarrow \int_1^2 \frac{1}{u^3}\,du = \left[\frac{-1}{2u^2}\right]_1^2 = \frac{3}{8} \text{ Ans.}$$

## Definite Integral Of Symmetric Functions Over Symmetric Intervals:

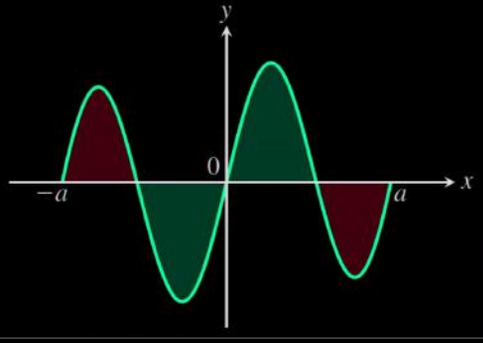
If $f(x)$ is even, $\int_{-a}^{a} f(x)\,dx = 2\int_0^a f(x)\,dx$

This can be clearly inferred from adjacent figure.




→ If $f(x)$ is an odd function,

$$\int_{-a}^{a} f(x)\,dx = 0 \quad \text{(See figure).}$$




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**Example:** Evaluate $\int_{-2}^{2} x^4 - 4x^2 + 6 \, dx$

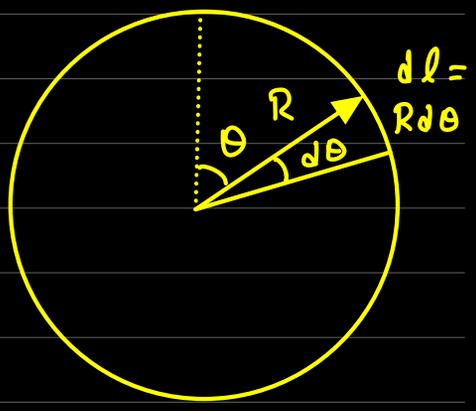
**Solution:** $f(-x) = f(x) \Rightarrow 2\int_{0}^{2} x^4 - 4x^2 + 6 = 2\left[\frac{x^5}{5} - \frac{4}{3}x^3 + 6x\right]_0^2$

$= 232/15$ Ans.

**Application Of Definite Integration:** As stated earlier, definite integration represents summation of infinitesimal parts, it can be used to evaluate perimeters, areas and volumes of various geometries. This will be clear from following examples.

**Example:** Using definite Integration show that circumference of a circle is $2\pi R$.

**Solution:** Consider a general element of the circle, located between angular positions $\theta$ and $\theta + d\theta$.




Now elemental length of this arc:

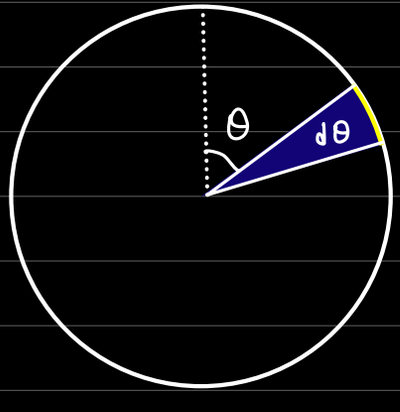
$$dl = R\,d\theta.$$

Now we need to add lengths of all such arcs lying from $\theta = 0$ to $\theta = 2\pi$. This is equivalent to performing a definite integral of the RHS. Thus we write:

$$\int_0^l dl = \int_0^{2\pi} R\,d\theta = R(\theta)_0^{2\pi} = 2\pi R \quad \text{QED.}$$

**Example:** Using Definite Integral prove that area of a circle is $\pi R^2$.

**Solution:** Consider an elemental sector of the circle lying between $\theta$ and $\theta + d\theta$. This can be treated like an isosceles triangle whose base is $R\,d\theta$ and height is $R$. Thus its elemental area:



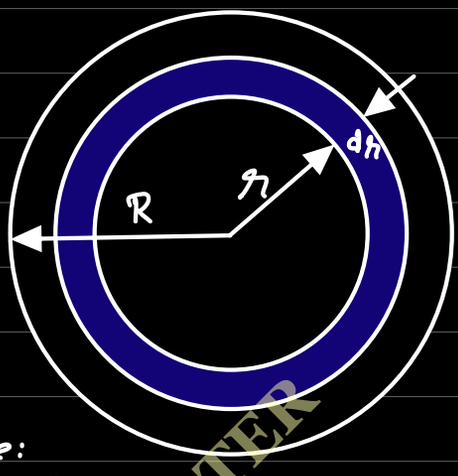

$$dA = \frac{1}{2} \times R\,d\theta \times R = \frac{1}{2} R^2 d\theta$$

Now areas of all such sectors needs to be added to get the area of complete circle. Thus :

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$$\int_0^A dA = \int_0^{2\pi} \frac{1}{2} R^2 \, d\theta = \frac{1}{2} R^2 [\theta]_0^{2\pi} = \pi R^2 \quad \text{thus} \quad A = \pi R^2 \quad \text{QED.}$$

**ALTERNATE METHOD 2:** Imagine an annular region of the circle lying between radii $r$ and $r+dr$. Elemental area of such an annulus:

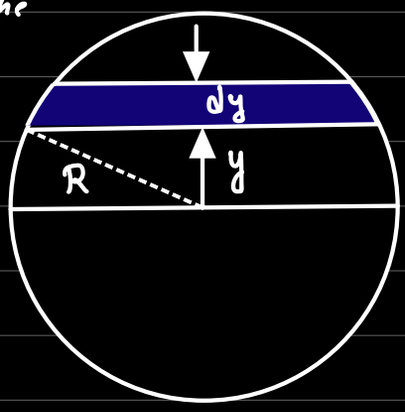



$$dA = 2\pi r \, dr$$

Now we need to add areas of all such Annuli between $r=0$ and $r=R$. Thus we integrate:

$$\int_0^A dA = \int_0^R 2\pi r \, dr = 2\pi \left[\frac{r^2}{2}\right]_0^R = \pi R^2 \quad \text{QED.}$$

**ALTERNATE METHOD 3:** Consider a strip element of the circle lying between coordinates $y$ and $y+dy$. The elemental area of the strip:




$$dA = 2\sqrt{R^2 - y^2}\, dy$$

Now areas of all such strips needs to be added from $y=-R$ to $y=+R$. Thus total area is evaluated using following Integration:

$$\int dA = \int_{-R}^{R} 2\sqrt{R^2 - y^2}\, dy = 4\int_0^R \sqrt{R^2 - y^2}\, dy \quad (\because \text{Even function}).$$

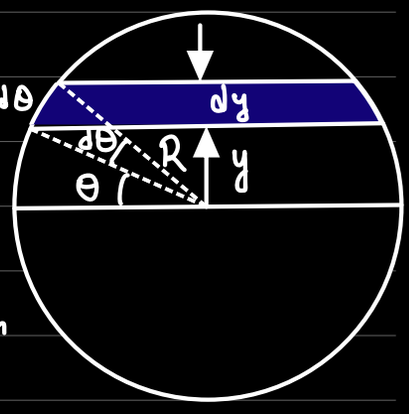
Now using $y = R\sin\theta$, $dy = R\cos\theta \, d\theta$, Now when $y = R$, $\theta = \pi/2$ Thus:

$$A = 4\int_0^{\pi/2} R^2 \cos^2\theta \, d\theta = 4R^2 \int_0^{\pi/2} \frac{\cos(2\theta)+1}{2} \, d\theta$$

$$= 2R^2 \left[\frac{\sin 2\theta}{2} + \theta\right]_0^{\pi/2} = \pi R^2 \quad \text{QED.}$$

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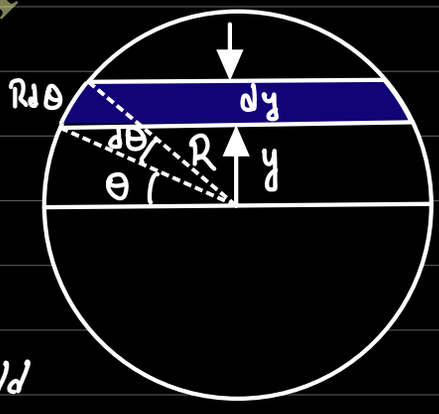
**EXAMPLE:** Using Definite Integral, prove that surface area of a spherical shell is $4\pi R^2$.




**Solution:** Consider a ring of the spherical shell lying between "latitudes" $\theta$ and $\theta+d\theta$. Elemental area of the ring: $dA = 2\pi R\cos\theta \times R\,d\theta$
(∵ slant height is $R\,d\theta$ and radius of ring is $R\cos\theta$).
Now we need to add areas of all such rings from $\theta = -\pi/2$ to $+\pi/2$ i.e "South Pole" to "North Pole".
Thus:

$$A = 2\pi R^2 \int_{-\pi/2}^{+\pi/2} \cos\theta\, d\theta = 4\pi R^2 \int_0^{\pi/2} \cos\theta\, d\theta = 4\pi R^2 \quad \text{QED.}$$

**EXAMPLE:** Using definite Integral prove that Volume of a solid sphere is $4/3\,\pi R^3$?

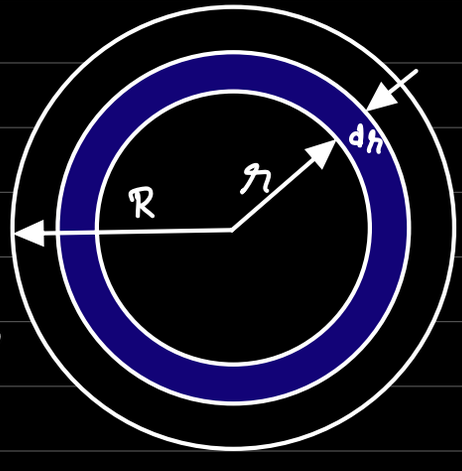



Solution: Consider a disc element of the sphere between vertical coordinates $y$ and $y+dy$.
Elemental volume of the disc is given as:

$$dV = \pi\left(\sqrt{R^2-y^2}\right)^2 dy$$, Now we need to add volumes of all such discs between $y=-R$ and $y=+R$. Thus,

$$V = \int_{-R}^{R} \pi(R^2-y^2)\,dy = 2\pi\left(R^2y - y^3/3\right)_0^R = 4/3\,\pi R^3 \quad \text{QED.}$$

**METHOD 2:** Consider a thin shell element between radii $r$ and $r+dr$. The elemental volume of the shell, $dV = 4\pi r^2 dr$, now we need to add the volumes of all such shells from $r=0$ to $r=R$.



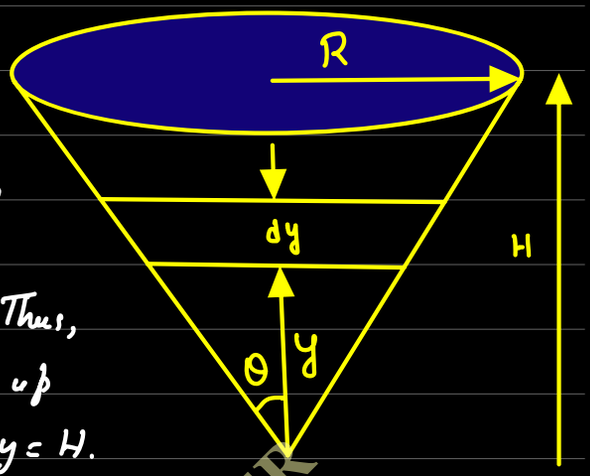

Now Integrating:

$$V = \int_0^R 4\pi r^2 dr = 4\pi\left[r^3/3\right]_0^R$$

$$\Rightarrow \quad V = 4/3\,\pi R^3 \quad \text{QED.}$$

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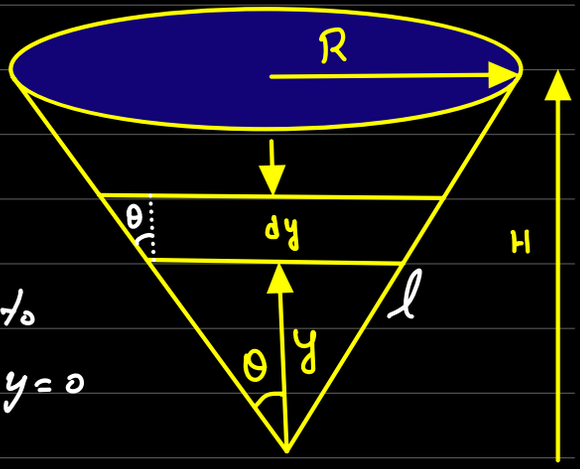
**EXAMPLE:** Using definite Integral prove that volume of a cone is $\frac{1}{3}\pi R^2 H$.




**Solution:** Consider elemental volume of the cone between vertical coordinates $y$ and $y+dy$.
The elemental volume of the element is given as: $dV = \pi (y\tan\theta)^2 . dy$ ($\because$ radius of the disc element is $y\tan\theta$) also $\tan\theta = R/H$. Thus, $dV = \pi y^2 \times (R^2/H^2)\, dy$, Now we need to add up volumes of all such disks between $y=0$ and $y=H$.
Thus integrating: $V = \frac{\pi R^2}{H^2}\int_0^H y^2 dy = \frac{1}{3}\pi R^2 H$ QED.

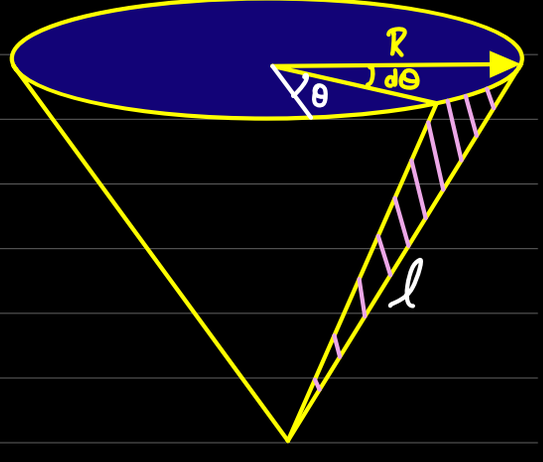
**EXAMPLE:** Using definite Integral prove that curved surface area of a cone is given as $S = \pi R l$ ($l$ is the slant height).

**Solution:** Consider a cap shaped element of the conical shell between heights $y$ and $y+dy$.




The area of the elemental cap is given as:
$dA = 2\pi y\tan\theta \times dy \sec\theta$
Now $\tan\theta = R/H$ and $\sec\theta = l/H$, thus,
$dA = 2\pi y\,(R/H)(l/H)\,dy$. Now we need to add up the areas of all such elements between $y=0$ to $y=H$. Now integrating:
$$A = \frac{2\pi l R}{H^2}\int_0^H y\,dy = \pi R L \quad \text{QED.}$$

**METHOD 2:** Consider an isosceles triangular elemental strip along the slant height of the cone.



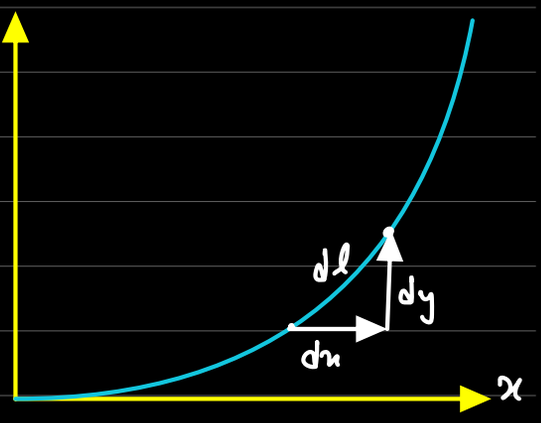

The area of the elemental strip:
$dA = \frac{1}{2} \times R d\theta \times l$, Now Integrating:
$A = \frac{1}{2} R l \int_0^{2\pi} d\theta = \pi R l$ QED.

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## Arc Length Of A Curve:

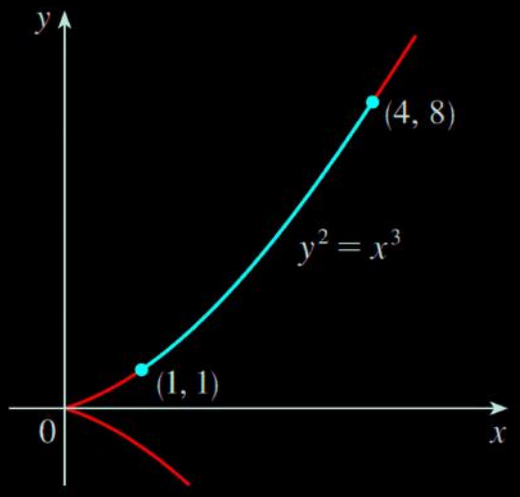
Consider an infinitesimal arc length of the curve $y = f(x)$ as shown in the figure using pythagoras theorem, $dl = \sqrt{(dx)^2 + (dy)^2}$

$$\Rightarrow \quad dl = dx\sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$ , Now to find total arc length between $x = a$ and $x = b$, we need to add up all such arc lengths. Thus:




$$\boxed{l = \int_a^b \sqrt{1 + y'^2}\, dx}$$

**Example:** Find the arc length of the curve defined by $y^2 = x^3$ between points (1,1) and (4,8).




**Solution:** $y = x^{3/2} \Rightarrow y' = \frac{3}{2}\sqrt{x}$

$\Rightarrow$ Required curve length $l = \int_1^4 \sqrt{1 + \frac{9}{4}x}$

$$\frac{4}{9} \times \frac{2}{3}\left[\left(1 + \frac{9}{4}x\right)^{3/2}\right]_1^4 = \frac{8}{27}\left[10^{3/2} - \left(\frac{13}{4}\right)^{3/2}\right]$$

$$= \frac{1}{27}\left[80\sqrt{10} - 13\sqrt{13}\right] \text{ Ans.}$$

## Integration By Parts:

Recall the product rule of differentiation:

$$\frac{d}{dx}\left[f(x)\, g(x)\right] = f(x)\, g'(x) + g(x)\, f'(x)$$

$$\Rightarrow \quad \int \left(f(x)\, g'(x) + g(x)\, f'(x)\right) dx = f(x)\, g(x)$$

$$\Rightarrow \quad \int f(x)\, g'(x)\, dx = f(x)\, g(x) - \int g(x)\, f'(x)\, dx$$

Now suppose $g'(x) = h(x)$ then:

$$\int f(x)\, h(x)\, dx = f(x)\int h(x)\, dx - \int f'(x)\left[\int h(x)\, dx\right] dx$$

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To remember simply:

$$\int uv = u\int v - \int [u' \int v]$$

**EXAMPLE:** Find $\int x \sin(x)\, dx$

**Solution:**

$u = x$, $v = \sin(x)$ $\Rightarrow$ required integral: $x[-\cos x] - \int 1(-\cos x)\, dx$

$= -x\cos x + \sin x + C$ Ans.

**EXAMPLE:** Find $\int \ln(x)\, dx$

Solution: Let us try $u = \ln(x)$, $v = 1$ then:

$\int uv = \ln(x) \times x - \int \frac{1}{x} \cdot x\, dx = x\ln(x) - x + C$ Ans.

**EXAMPLE:** Find $\int t^2 e^t dt$

**Solution:** $u = t^2$, $v = e^t$ $\Rightarrow \int uv = t^2 e^t - \int 2t\, e^t dt$

Now for evaluating $\int t e^t dt$ again use by parts with $u = t$, $v = e^t$ thus $\int te^t = te^t - \int e^t dt = te^t - e^t$, thus our require integration:

$$t^2 e^t - 2[te^t - e^t] = e^t[t^2 - 2t + 2] + C \text{ Ans.}$$

**EXAMPLE:** Evaluate $\int e^x \sin x\, dx$

**Solution:** Let $I = \int e^x \sin x\, dx$, $u = e^x$, $v = \sin(x)$ $\Rightarrow$

$I = e^x[-\cos x] - \int e^x[-\cos x]\, dx = -e^x\cos x + \int e^x \cos x\, dx$

Now again let $u = e^x$, $v = \cos x$ for solving the integration part then,

$\int e^x \cos x\, dx = e^x \sin x - \int e^x \sin(x)\, dx = e^x \sin(x) - I$ thus:

$I = -e^x\cos x + e^x \sin(x) - I \Rightarrow I = \frac{e^x[\sin x - \cos x]}{2} + C$

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## ILATE RULE FOR INTEGRATION BY PARTS:

| Letters of ILATE Rule | Abbreviations | Examples |
|---|---|---|
| I | Inverse trigonometric functions | $\sin^{-1}x$, $\cos^{-1}x$, etc |
| L | Logarithmic functions | $\log x$, $\ln x$ |
| A | Algebraic functions | $x^2$, $\sqrt{x}$ |
| T | Trigonometric functions | $\sin x$, $\cos x$ |
| E | Exponential functions | $e^x$, $2^x$ |

While performing integration by parts, we have to decide which function to consider as $u$ and which one to put as $v$. For most cases, such integrations are simplified by priority rule for $u$ called ILATE rule. According to this, $u$ must be chosen to be inverse trigonometric, logarithmic, algebraic, trigonometric and exponential in that order. For instance suppose we wish to integrate $x^2 \ln(x)$ then $u$ should be taken as $\ln(x)$ and not $x^2$ because logarithmic functions have higher priority over algebraic ones.

## LINEAR DIFFERENTIAL EQUATIONS:

Any equation of the form: $y' + p(x)y = q(x)$ is called a first order linear differential equation. This type of equations are solved using integrating factor $I = e^{\int p\,dx}$. Multiplying by the integrating factor makes the LHS a perfect differential and then the equation becomes variable reparable. This will be more clear from following example:

## EXAMPLE:

Solve the D.E.: $y' + \frac{1}{x}y = 2$

**Solution:** Here by comparison with standard linear equation, $p(x) = 1/x$. Thus integrating factor, $I = e^{\int \frac{1}{x}dx} = e^{\ln x} = x$. Now multiplying entire equation by integrating factor:

$$xy' + y = 2x \quad \text{or} \quad \frac{d}{dx}(xy) = 2x$$

(See LHS is perfect differential of $yI$), Now taking $x$ to the other side and integrating:

$$xy = \int 2x\,dx \Rightarrow xy = x^2 + C$$

$$\Rightarrow y = (x^2 + C)/x \text{ Ans.}$$

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→ Thus general solution to the linear form is:

$$y = \frac{1}{I}\int I\, q(x)\, dx \quad \text{where} \quad I = e^{\int p\, dx}$$

**EXAMPLE:** Solve the differential equation: $dy/dx + 3x^2 y = 6x^2$

**Solution:** $p(x) = 3x^2$ $q(x) = 6x^2$, $I = e^{\int 3x^2 dx} = e^{x^3}$

Thus required solution:

$$y = \frac{1}{e^{x^3}} \int 6x^2 e^{x^3} dx$$

Now substitute $u = x^3 \Rightarrow 3x^2 dx = du$ thus integration becomes $2\int e^u du$

$= 2e^{x^3}$, thus $y = \frac{1}{e^{x^3}}[2e^{x^3} + c]$, thus $y = 2 + ce^{-x^3}$ Ans.

**EXAMPLE:** Solve the initial value problem: $x^2 y' + xy = 1$, $x > 0$, $y(1) = 2$

**Solution:** Dividing by $x^2$, $y' + \frac{1}{x} y = \frac{1}{x^2}$, $p = 1/x$, $q = 1/x^2$,

$I(x) = e^{\int p\, dx} = e^{\ln x} = x \Rightarrow y = \frac{1}{x}\int x \times \frac{1}{x^2} dx$

$\Rightarrow y = \frac{1}{x}[\ln|x| + c]$, Now $y(1) = 2 \Rightarrow \frac{1}{1}(\ln(1) + c) = 2$

$\Rightarrow c = 2$

Thus required solution: $y = \frac{1}{x}[\ln|x| + 2]$ Ans.

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